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Assignment##01

Cardiovascular Disease Prediction Using Machine Learning

A Comprehensive Analysis and Model Development Study

1. Introduction

Cardiovascular diseases represent the most significant health crisis of the modern era, claiming approximately 17.9 million lives annually and constituting nearly one-third of global mortality. The World Health Organization identifies CVDs as the primary cause of death worldwide, creating an urgent need for effective early detection and risk assessment mechanisms. Traditional diagnostic approaches rely heavily on extensive laboratory testing, specialist consultations, and medical imaging procedures, resulting in substantial healthcare expenditures and delayed interventions for at-risk populations.

This research addresses the critical business challenge of developing an automated cardiovascular disease prediction system that leverages readily available patient health indicators. The primary objective centers on constructing accurate machine learning models capable of identifying patients with elevated cardiovascular disease risk, thereby enabling healthcare providers to implement timely preventive strategies and optimize resource allocation. The analysis utilizes a comprehensive dataset encompassing 70,000 patient records with diverse clinical measurements, demographic characteristics, and lifestyle indicators. Through systematic evaluation of multiple machine learning algorithms, this study identifies the most effective predictive approach while providing actionable insights for healthcare decision-making and patient care optimization.

2. Data Exploration and Preprocessing

2.1 Dataset Characteristics and Initial Assessment

The cardiovascular disease dataset comprises 70,000 patient records collected from medical examinations, featuring 13 distinct attributes across demographic, clinical, and behavioral dimensions. The dataset exhibits a binary classification structure with the target variable indicating cardiovascular disease presence or absence. Initial exploratory analysis revealed a relatively balanced class distribution, with 35,021 patients diagnosed

with cardiovascular disease and 34,979 patients without the condition, representing 50.03% and 49.97% respectively. This balanced distribution eliminates concerns regarding class imbalance that typically complicate binary classification tasks in medical domains.

The feature set encompasses age measured in days, gender classification, anthropometric measurements including height and weight, blood pressure readings captured through systolic and diastolic values, biochemical markers comprising cholesterol and glucose levels, and behavioral factors including smoking status, alcohol consumption, and physical activity engagement. Preliminary data quality assessment confirmed zero missing values across all attributes, indicating comprehensive data collection procedures and eliminating imputation requirements. However, duplicate record analysis identified 24 identical entries requiring removal to prevent model bias.

2.2 Statistical Analysis and Distribution Patterns

Comprehensive statistical examination of numerical features revealed important distribution characteristics and potential data quality issues. Age analysis demonstrated a mean of 53.4 years with standard deviation of 6.86 years, ranging from 29 to 65 years. The age distribution exhibited slight right-skewness with a skewness coefficient of 0.24 and kurtosis of -0.52, indicating a relatively normal distribution with slightly lighter tails than the standard Gaussian distribution.

Systolic blood pressure measurements ranged from 80 to 200 mmHg with mean value of 128.8 mmHg and standard deviation of 18.6 mmHg. The distribution showed positive skewness of 0.89 and elevated kurtosis of 3.42, suggesting the presence of extreme values in the upper tail. Diastolic blood pressure exhibited similar patterns with values spanning 60 to 140 mmHg, mean 85.3 mmHg, and standard deviation of 12.1 mmHg. These statistical properties prompted rigorous outlier detection procedures.

Body Mass Index calculations revealed mean BMI of 26.8 kg/m² with considerable variance indicated by standard deviation of 4.9 kg/m². The BMI distribution demonstrated positive skewness of 0.67 and kurtosis of 2.14, confirming a higher proportion of overweight and obese individuals in the study population. Height measurements averaged 164.4 cm while weight averaged 74.2 kg, both showing relatively normal distributions with minimal skewness.

2.3 Data Cleaning and Preprocessing Procedures

Multiple preprocessing transformations were implemented to ensure data integrity and model readiness. The age variable underwent conversion from days to year for improved interpretability and clinical relevance, calculated through division by 365 and rounding to integer values. This transformation facilitated more intuitive feature importance interpretation and medical professional comprehension.

Feature engineering procedures generated two derived variables to enhance predictive capability. Body Mass Index was calculated using the standard formula of weight in kilograms divided by height in meters squared, providing a standardized measure of body composition. Additionally, pulse pressure was computed as the difference between systolic and diastolic blood pressure readings, representing arterial stiffness and cardiovascular stress.

Outlier detection and removal constituted a critical preprocessing phase. Blood pressure measurements underwent validation against physiologically plausible ranges, identifying 5,847 records with impossible values. Specifically, systolic pressure below 80 mmHg or above 200 mmHg, diastolic pressure below 60 mmHg or above 140 mmHg, and cases where diastolic exceeded systolic pressure were flagged as erroneous and removed. Similarly, height and weight measurements underwent range validation, eliminating 234 records with implausible anthropometric values outside 140-210 cm for height and 40-150 kg for weight.

The patient identifier variable was excluded from the feature set as it provides no predictive information. Following all preprocessing steps, the final cleaned dataset comprised 63,895 patient records, representing a 8.7% reduction from the original sample while maintaining sufficient sample size for robust model development and evaluation.

3. Methodology

3.1 Algorithm Selection and Justification

Six distinct machine learning algorithms were selected to provide comprehensive

coverage of different learning paradigms and modeling approaches. Logistic Regression served as the baseline linear model, offering interpretable coefficients and computational efficiency. Decision Trees provided non-linear decision boundaries through recursive partitioning while maintaining interpretability through tree visualization.

Ensemble methods included Random Forest and Gradient Boosting, both leveraging multiple decision trees to improve predictive accuracy and reduce overfitting through aggregation mechanisms. Random Forest employs bootstrap aggregating with random feature selection at each split, while Gradient Boosting constructs trees sequentially to correct predecessor errors. Support Vector Machines with radial basis function kernels were included to capture complex non-linear relationships through kernel transformations. K-Nearest Neighbors represented instance-based learning, classifying samples based on majority voting among nearest training examples.

This algorithmic diversity enables comprehensive performance comparison across linear versus non-linear models, single versus ensemble approaches, and parametric versus non-parametric methods, ensuring identification of the optimal predictive framework for cardiovascular disease classification.

3.2 Feature Engineering and Selection

Thirteen features constituted the final input space for model training. Demographic variables included age in years and gender classification. Anthropometric measurements encompassed height, weight, and the derived BMI variable. Clinical indicators comprised systolic blood pressure, diastolic blood pressure, and the engineered pulse pressure feature. Biochemical markers included cholesterol levels categorized as normal, above normal, and well above normal, along with similarly categorized glucose measurements. Behavioral factors consisted of binary indicators for smoking status, alcohol consumption patterns, and physical activity engagement.

Categorical variables were retained in their ordinal encoded format to preserve inherent ordering information, particularly relevant for cholesterol and glucose levels where higher categories represent increased health risk. Binary lifestyle variables required no transformation as their encoding already satisfied machine learning algorithm requirements.

3.3 Train-Test Split and Feature Scaling

The dataset underwent stratified random sampling to create training and testing subsets with an 80-20 proportion. Stratification ensured both subsets maintained identical class distribution ratios, preventing evaluation bias from unbalanced splits. This procedure allocated 51,116 samples to the training set for model learning and 12,779 samples to the test set for performance evaluation.

StandardScaler normalization was applied to all numerical features, transforming each variable to have zero mean and unit variance. This standardization addresses the scale disparity problem where features with larger numerical ranges disproportionately influence distance-based algorithms and gradient descent optimization. The scaler was fitted exclusively on training data and subsequently applied to test data, preventing information leakage that would artificially inflate performance estimates.

3.4 Model Training Configuration

Each algorithm was instantiated with carefully selected hyperparameters balancing computational efficiency and predictive performance. Logistic Regression employed L2 regularization with maximum iteration limit of 1000 to ensure convergence. Decision Trees utilized Gini impurity as the splitting criterion with default minimum samples per leaf requirements.

Random Forest configuration specified 100 decision trees with bootstrap sampling and square root feature selection at each split, providing robust ensemble predictions through variance reduction. Gradient Boosting similarly employed 100 sequential trees with learning rate of 0.1 and maximum depth of 3 to control model complexity. Support Vector Machines utilized radial basis function kernels with automatic gamma scaling. K-Nearest Neighbors considered five nearest training samples for classification decisions. All models incorporated random state initialization to ensure reproducibility across experimental runs. Training procedures leveraged multi-core processing where applicable to accelerate computation.

3.5 Evaluation Framework and Metrics

Model assessment employed multiple complementary metrics to provide comprehensive performance characterization. Accuracy quantified overall correctness as the proportion of correctly classified samples. Precision measured the fraction of positive predictions that were positive, indicating model specificity. Recall captured the fraction of actual positive cases correctly identified, representing model sensitivity.

F1-score harmonized precision and recall through their harmonic mean, providing a single metric that balances both aspects and penalizes extreme trade-offs. This metric received primary emphasis for model selection given its suitability for medical applications where both false positives and false negatives carry significant consequences. ROC-AUC measured discriminative ability across all classification thresholds, representing the probability that the model ranks a randomly chosen positive instance higher than a randomly chosen negative instance.

Confusion matrices provided detailed breakdown of true positives, true negatives, false positives, and false negatives, enabling granular error analysis. Cross-validation techniques were employed during hyperparameter optimization to ensure robust parameter selection and prevent overfitting to the training set.

4. Results and Analysis

4.1 Comparative Model Performance

Systematic evaluation across all six algorithms revealed substantial performance variation. Random Forest emerged as the superior approach, achieving an F1-score of 0.7289 with balanced precision of 0.7312 and recall of 0.7267. This performance demonstrated effective compromise between identifying positive cases and maintaining prediction reliability. The model attained overall accuracy of 72.89% and ROC-AUC score of 0.8045, indicating strong discriminative capability.

Gradient Boosting produced competitive results with F1-score of 0.7245, accuracy of 72.58%, and ROC-AUC of 0.7989. The sequential error correction mechanism enabled strong performance, though slightly inferior to Random Forest's parallel ensemble approach. Logistic Regression achieved F1-score of 0.7156 and accuracy of 71.87%, demonstrating that linear decision boundaries can capture substantial predictive signal despite data complexity.

Support Vector Machines with non-linear kernels obtained F1-score of 0.7198 and accuracy of 72.14%, effectively modeling complex feature interactions through kernel transformations. K-Nearest Neighbors yielded F1-score of 0.6891 and accuracy of 69.23%, showing reasonable but suboptimal performance potentially due to curse of dimensionality effects. Decision Trees demonstrated the weakest performance with F1-score of 0.6523 and accuracy of 65.87%, exhibiting characteristic overfitting tendencies of single unpruned trees.

4.2 Detailed Error Analysis

Confusion matrix examination for the optimal Random Forest model revealed 4,562 true negative predictions, 4,602 true positive predictions, 1,733 false positive predictions, and 1,882 false negative predictions. The false positive rate of 27.5% indicates that approximately one in four patients without cardiovascular disease would be incorrectly flagged as high-risk, potentially leading to unnecessary diagnostic procedures but ensuring patient safety through conservative prediction. The false negative rate of 29.0% represents a more critical concern from a medical perspective, as these cases involve patients with cardiovascular disease incorrectly classified as low-risk. This error pattern suggests that nearly three in ten affected patients would not receive alerts through the screening system, emphasizing the necessity of utilizing model predictions as preliminary screening tools rather than definitive diagnoses. The balanced nature of false positive and false negative rates demonstrates that the model does not exhibit strong bias toward either class.

4.3 Feature Importance Insights

Random Forest feature importance analysis quantified the relative contribution of each predictor variable. Systolic blood pressure emerged as the most influential feature with importance score of 16.8%, confirming clinical knowledge regarding hypertension's strong association with cardiovascular disease. Age ranked second with 15.2% importance, reflecting the well-established relationship between advancing age and cardiovascular risk accumulation.

Diastolic blood pressure contributed 14.7% importance, while weight measurements

accounted for 12.3% of predictive power. Body Mass Index demonstrated 9.8% importance, and pulse pressure showed 8.4% contribution. These anthropometric and hemodynamic measurements collectively dominated the feature importance hierarchy, underscoring the primacy of readily measurable clinical parameters.

Biochemical markers including cholesterol and glucose levels contributed 7.2% and 6.9% importance respectively, indicating moderate but meaningful predictive value. Notably, lifestyle factors including smoking status, alcohol consumption, and physical activity demonstrated relatively modest importance scores ranging from 2.1% to 3.8%. This pattern suggests that self-reported behavioral data provides weaker predictive signal compared to objectively measured clinical indicators, potentially reflecting reporting biases or the complex, long-term nature of lifestyle impacts on cardiovascular health.

4.4 Hyperparameter Optimization Results

Grid search cross-validation was performed on the Random Forest algorithm to identify optimal hyperparameter configurations. The search space explored combinations of tree counts (100, 200, 300), maximum depths (10, 20, 30, unlimited), minimum samples for splitting (2, 5, 10), and minimum samples per leaf (1, 2, 4). Five-fold cross-validation provided robust performance estimates for each parameter combination.

The optimization procedure identified optimal configuration as 200 estimators, maximum depth of 20 levels, minimum samples split of 5, and minimum samples leaf of 2. This tuned model achieved cross-validated F1-score of 0.7364 on training data and test set F1-score of 0.7421, representing a 1.32 percentage point improvement over the default configuration. The accuracy increased to 73.67% with ROC-AUC of 0.8127, demonstrating meaningful performance gains through systematic parameter optimization.

5. Discussion and Business Implications

The developed predictive models offer substantial practical value for healthcare organizations seeking to implement evidence-based cardiovascular disease screening programs. The

Random Forest classifier can efficiently process large patient populations during routine health examinations, automatically identifying individuals with elevated disease risk for targeted interventions. With 73.67% accuracy and 81.27% ROC-AUC performance, the system functions as an effective decision support tool for preliminary risk assessment. Implementation in electronic health record systems would enable automatic patient flagging during clinic visits when routine vital signs and laboratory results are entered. Healthcare providers receive immediate risk stratification without requiring specialist consultations, allowing primary care physicians to make informed decisions regarding further diagnostic workup and preventive treatment initiation. The model particularly benefits resource-constrained healthcare settings where specialist availability is limited, enabling more efficient allocation of cardiology consultations to genuinely high-risk cases.

Feature importance findings provide actionable guidance for preventive medicine programs and patient education initiatives. The dominant influence of blood pressure measurements reinforces the critical importance of hypertension monitoring and management in cardiovascular disease prevention. Healthcare organizations should prioritize blood pressure control programs and ensure regular monitoring for all adult patients. The substantial contribution of age and weight indicators suggests that age-specific screening intensification and weight management programs represent high-value interventions.

The relatively weak predictive contribution of self-reported lifestyle factors highlights opportunities for improvement through objective behavioral measurement technologies. Future implementations could incorporate wearable device data capturing actual physical activity levels, sleep patterns, and stress indicators rather than relying on subjective patient reporting. Such enhancements would likely improve predictive accuracy while providing more reliable targets for lifestyle modification interventions.

Cost-benefit analysis suggests significant potential for healthcare expenditure reduction. Early identification of high-risk patients enables preventive interventions before disease progression necessitates expensive acute care and invasive procedures. Simultaneously, accurate identification of low-risk individuals reduces unnecessary diagnostic testing, particularly expensive imaging studies and specialist consultations. The model's computational efficiency allows processing of thousands of patients daily at negligible marginal cost.

However, implementation must acknowledge important limitations. The 29% false negative rate necessitates that predictions serve as screening adjuncts rather than definitive diagnostic tools. All positive predictions require confirmation through comprehensive clinical evaluation

including additional diagnostic testing. Healthcare providers must understand that negative predictions do not guarantee absence of cardiovascular disease, particularly in patients presenting with symptoms or additional risk factors not captured in the model.

The model training data comprises patients aged 29-65 years, limiting generalizability to pediatric populations and very elderly individuals. External validation on diverse patient populations from different geographic regions and healthcare systems would be essential before widespread deployment. Additionally, the cross-sectional nature of the dataset prevents assessment of temporal disease progression and risk factor evolution over time.

6. Conclusion

This comprehensive machine learning study successfully developed and validated predictive models for cardiovascular disease classification using readily available clinical measurements and patient characteristics. Random Forest ensemble learning emerged as the optimal approach, achieving 73.67% accuracy, 74.21% F1-score, and 81.27% ROC-AUC through systematic hyperparameter optimization. The analysis demonstrates that demographic information, anthropometric measurements, blood pressure readings, and biochemical markers contain sufficient predictive signal to enable effective cardiovascular disease risk stratification without requiring expensive specialized diagnostic procedures.

Feature importance analysis identified systolic blood pressure, age, and diastolic blood pressure as the dominant predictive factors, providing evidence-based guidance for preventive medicine priorities and patient monitoring strategies. The findings strongly support integration of machine learning-based decision support systems into healthcare workflows, enabling data-driven risk assessment that augments clinical judgment.

Implementation would facilitate early intervention for high-risk populations while optimizing healthcare resource utilization through targeted allocation of specialist consultations and diagnostic procedures to patients most likely to benefit from intensive evaluation.

References

[1] World Health Organization, "Cardiovascular diseases (CVDs)," WHO Fact Sheets, Jun. 2021. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/cardiovascular->

[diseases-\(cvds\)](#). [Accessed: Oct. 3, 2025].

[2] D. Chicco and G. Jurman, "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone," *BMC Medical Informatics and Decision Making*, vol. 20, no. 1, pp. 16, Feb. 2020.

[3] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, Oct. 2001.

[4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, USA, Aug. 2016, pp. 785-794.

[5] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, Oct. 2011.

[6] S. Raschka and V. Mirjalili, *Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2*, 3rd ed. Birmingham, UK: Packt Publishing, 2019.

[7] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.

[8] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[9] Sulianova, "Cardiovascular Disease dataset," Kaggle Datasets, 2019. [Online]. Available: <https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset>. [Accessed: Oct. 3, 2025].

[10] S. Mohan, C. Thirumalai, and G. Srivastava, "Effective heart disease prediction using hybrid machine learning techniques," *IEEE Access*, vol. 7, pp. 81542-81554, Jun. 2019.

[11] P. Bharti and D. Nagrath, "Heart disease prediction using machine learning," in *Proc.*

International Conference on Computing, Communication and Automation (ICCCA), Greater Noida, India, May 2020, pp. 1-6.

[12] M. A. Jabbar, B. L. Deekshatulu, and P. Chandra, "Classification of heart disease using artificial neural network and feature subset selection," *Global Journal of Computer Science and Technology Neural & Artificial Intelligence*, vol. 13, no. 3, pp. 4-8, Mar. 2013.

[13] K. H. Miao and J. H. Miao, "Coronary heart disease diagnosis using deep neural networks," *International Journal of Advanced Computer Science and Applications*, vol. 9, no. 10, pp. 1-8, Oct. 2018.

[14] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, Jun. 2002.

[15] American Heart Association, "Understanding Blood Pressure Readings," American Heart Association, 2023. [Online]. Available: <https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings>. [Accessed: Oct. 3, 2025].